DNA/Protein Sequence Analysis from UniProt -CodeApha Internship

Bioinformatics

Introduction

A protein sequence in FASTA format was obtained from the UniProt database. The sequence corresponds to a protein of interest and was used as the query for sequence similarity analysis.

Methodology

The FASTA protein sequence was submitted to the NCBI BLAST tool using the BLASTp program, which compares a protein query sequence against a protein database.

The aim was to identify homologous protein sequences and evaluate their similarity to the query sequence.

Results

The BLAST search returned multiple significant matches, primarily corresponding to insulin proteins from different organisms. The top matches were obtained from closely related primates.

Examples of top hits include:

1. **insulin preproprotein [Pan troglodytes]**
    - Score: 221 bits
    - E-value: 1e-72
    - Identity: 108/110 (98%)
    - Positives: 109/110 (99%)

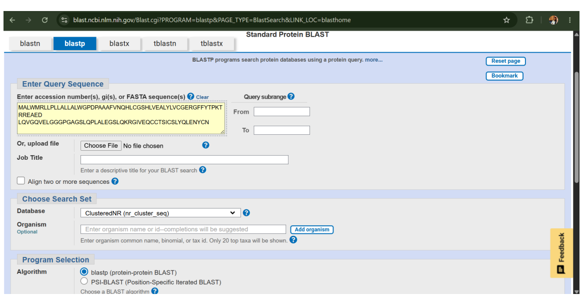
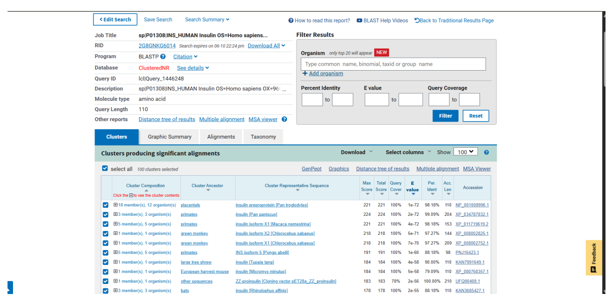
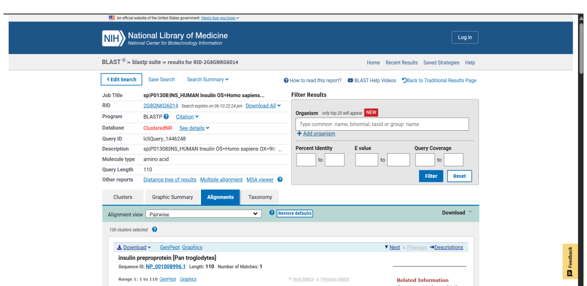
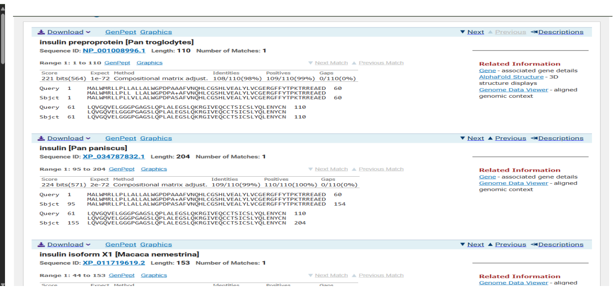
2. **Insulin [Pan paniscus]**
    - Score: 224 bits
    - E-value: 2e-72
    - Identity: 109/110 (99%)
    - Posities: 110/110 (100%)

3. **insulin isoform X1 [Macaca nemestrina]**
    - Score: 221 bits
    - E-value: 4e-72
    - Identity: 108/110(98%)
    - Positives: 108/110(98%)

4. **insulin isoform X2 [Chlorocebus sabaeus]**
    - Score: 218 bits
    - E-value: 56-71
    - Identity: 107/110(97%)
    - Positives: 107/110(97%)

5. **insulin isoform X1 [Chlorocebus sabaeus]**
    - Score: 218 bits
    - E-value: 7e-70
    - Identity: 107/110(97%)
    - Positives: 107/110(97%)

All matches showed very high query coverage (~100%), high alignment scores, and extremely low E-values, indicating highly significant alignments.

Analysis and Interpretation

The BLAST result shows the query sequence is highly similar to insulin proteins found in various species, particularly primates. The percentage identity ranges from 98% to 100%, while similarity (positives) is nearly 100% across all top matches.

The extremely low E-values (on the order of 10^{-72}) indicate that these matches are not due to random chance but represent true biological relationships.

The high degree of similarity across different species suggests that this protein is evolutionarily conserved. Conserved proteins typically perform essential biological functions that are critical for survival. In this case, insulin is a key hormone involved in the regulation of blood glucose levels.

Based on these findings, the query sequence can be confidently identified as insulin or a closely related protein. The conservation of this sequence across species highlights its functional importance in metabolic processes.

Conclusion

The BLAST analysis successfully identified homologous sequences corresponding to insulin proteins across multiple organisms. The high identity, strong alignment scores, and low E-values confirm that the query sequence is highly conserved and biologically significant.

This analysis demonstrates the effectiveness of BLAST in identifying protein function and evolutionary relationships based on sequence similarity.